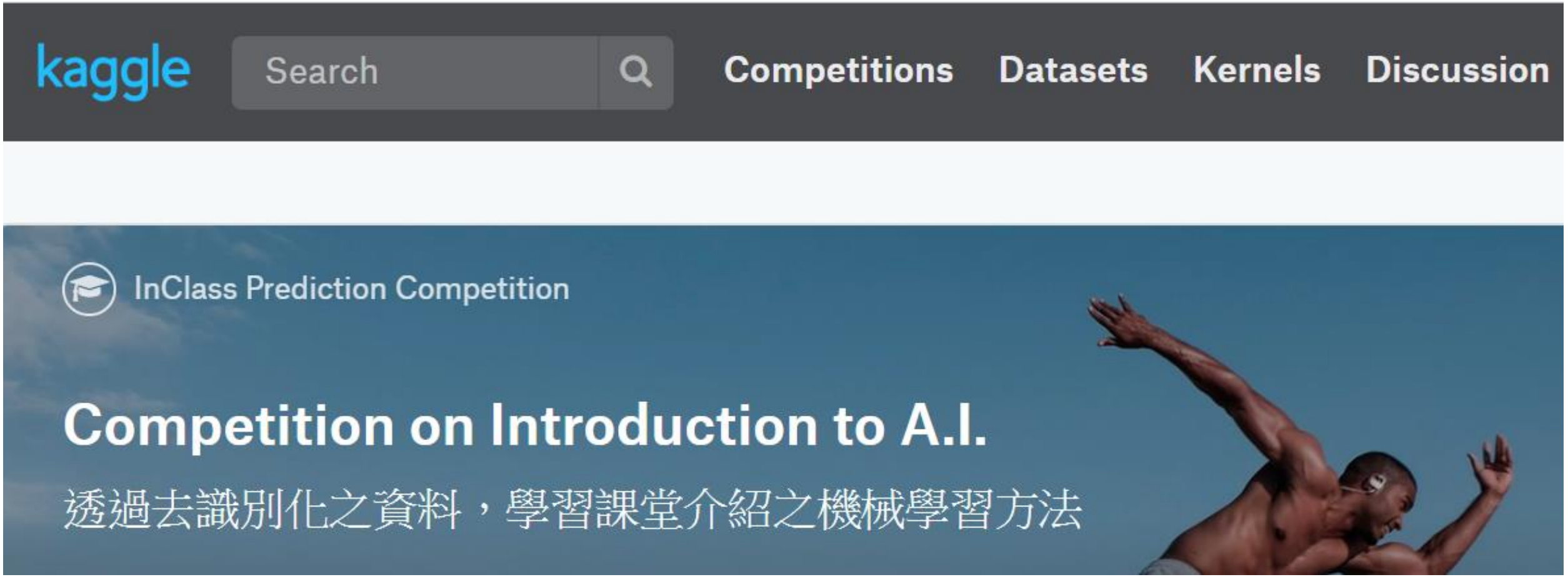
New dataset

https://www.kaggle.com/c/competition-on-introduction-to-ai



比賽目的:

練習在人工智慧導論課堂所學習到的資料分析及程式撰寫方法

比賽方式:

將解答上傳至kaggle，每天最多上傳2次

比賽成績:

預測準確度(accuracy)作為成績參考依據

比賽內容:

此資料為人體運動相關數據，詳細內容請見資料(Data)頁籤

Data Description

- **train.csv** - the training set
- **test.csv** - the test set
- **sampleSubmission.csv** - a sample submission file

此資料包含以下欄位

- id : 資料編號
- accel_body : 軀幹加速規讀數
- magnet_body : 軀幹磁力計讀數
- gyros_upper_limb : 上肢陀螺儀讀數
- accel_upper_limb : 上肢加速規讀數
- magnet_upper_limb : 上肢磁力計讀數
- gyros_body : 軀幹陀螺儀讀數
- gyros_lower_limb : 下肢陀螺儀讀數
- accel_lower_limb : 下肢加速規讀數
- magnet_lower_limb : 下肢磁力計讀數
- Category ; 運動類別

感測器種類

陀螺儀(gyroscope) : 是一種用來感測與維持方向的裝置，基於角動量守恆的理論設計出來的。

加速規 (accelerometer): 又稱加速計、加速針、加速度傳感器、重力加速度傳感器等等，是測量加速度的裝置。

磁力計 (magnetometer) : 輔助陀螺儀，彌補重力加速度對水平角度旋轉偵測誤差。

Data Sources

- data.zip
  - sampleSubmission.csv
  - test.csv
  - train.csv

train - Excel

檔案 常用 插入 版面配置 公式 資料 校閱 檢視 告訴我您想要執行的動作...

A1 id

	A	B	C	D	E	F	G	H	I	J	K
1	id	accel_body	magnet_body	gyros_upper_limb	accel_upper_limt	magnet_upper_lir	gyros_body	gyros_low	accel_lowe	magnet_lo	Category
2	0	-109.9939449		4.17E-06	1311.714258	very_fast	very_slow	on	on	off	0
3	1	79.96724561	23.64252469	93.4968092	-641.751055	very_slow	very_slow	off	on	on	1
4	2	-310.9951709	12.37888681	0.021659896	-45.94592181	very_fast	slow	on	off	on	0
5	3	-246.6944752	5.037199133	322.1115885	-412.6555543	very_fast	very_slow	off	off	off	1
6	4	21.75131312	21.25648253	0.005548748	-148.2052275	fast	very_slow	off	on	on	0
7	5	-159.5214974	32.10815904	0.686390776	-782.608156	medium	very_slow	off	on	on	0
8	6	-342.7025737	20.9631353	76.30372953	1445.748058	medium	very_slow	off	on	off	1
9	7	-25.07402055	5.346989571	780.2790363	274.0198011	very_slow	medium	on	off	on	4
10	8	-315.2850753	16.50293563	73.30118075	-661.2317301	fast	very_slow	off	on	on	0
11	9	-155.4296034	20.34494937	22.64935284	537.3357506	medium	slow	off	off	on	3
12	10	-471.6163621		109.9321315	294.9472839	very_slow	very_slow	off	on	on	1
13	11	-473.0200721		8.03E-06	211.3845351	very_fast	very_slow	on	on	on	0
14	12	301.9987129	18.99659466	235.9859123	196.846873	fast	slow	off	on	on	3
15	13	-86.03199311	28.97369688	0.119546749	-121.7587641	very_slow	very_fast	on	on	on	4
16	14	36.35536799		0.001553768	-254.0385161	very_fast	very_slow	on	on	on	0
17	15	-36.66520244		63.51170706	-591.2750414	slow	very_slow	on	on	on	0
18	16	-167.4516292	18.59164403	70.07552721	-380.5123703	very_fast	very_slow	off	on	on	1
19	17	192.8287699	7.709783851	43.58223336	298.2531695	very_fast	very_slow	off	on	on	1
20	18	-241.2675966	11.31435756	140.1605518	-1045.360517	fast	very_slow	off	on	off	1
21	19	-279.3180275	14.04786706	503.6542526	1169.248098	fast	slow	off	on	on	1
22	20	-188.3253662	26.74623422	6.024747473	1106.327759	very_fast	slow	off	off	off	0

sampleSubmission - Excel

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A1 id

	A	B	C	D	E	F	G	H	I	J
1	id	Category								
2	3219	4								
3	3220	0								
4	3221	4								
5	3222	1								
6	3223	4								
7	3224	3								
8	3225	4								
9	3226	0								
10	3227	1								

參考資料：

- OneHot encoding 請參考 Keras_01_Iris 檔案
- Missing data 請參考 Keras_03_Stock 檔案

Acknowledgement

- The REALDISP Activity Recognition Dataset Data Set was downloaded from UC Irvine Machine Learning Repository at https://archive.ics.uci.edu/ml/datasets/REALDISP+Activity+Recognition+Dataset#

- We thank to Banos at. el. providing valuable dataset. The dataset was modified and simplified for education purpose.

- References:

- Banos, O., Toth M. A., Damas, M., Pomares, H., Rojas, I. Dealing with the effects of sensor displacement in wearable activity recognition. Sensors vol. 14, no. 6, pp. 9995-10023 (2014).

- Banos, O., Toth, M. A., Damas, M., Pomares, H., Rojas, I., Amft, O. A benchmark dataset to evaluate sensor displacement in activity recognition. Proceedings of the 14th International Conference on Ubiquitous Computing (Ubicomp 2012), Pittsburgh, USA, September 5-8, (2012).